

Cross-Modal Semantic Alignment for Artwork Retrieval

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Abstract. This report studies semantic alignment between artworks and textual information using a retrieval-based evaluation on the SemArt dataset. It examines how visual and textual representations can be mapped into a shared semantic space, and how this mapping depends on both the representation model and the form of textual input.

To address this question, the report first compares different representation models in a text-to-image retrieval setting. Results show that CLIP ViT-B/32 outperforms CLIP RN50, supporting its use in subsequent experiments. The analysis then focuses on the role of textual information in retrieval performance.

Natural language descriptions, structured metadata, and their combinations are compared under easy and difficult retrieval conditions. The results indicate that descriptions provide strong semantic signals across settings, while adding metadata improves retrieval performance primarily in more difficult cases. Metadata alone does not support fine-grained semantic alignment.

Keywords: Cross-modal retrieval, Semantic alignment, Art understanding, Vision-language models, Multimodal representation

1 Introduction

Cross-modal retrieval aims to connect visual and textual information within a shared semantic space. In this setting, a model ranks images based on their semantic similarity to a given text query. Recent vision-language models have shown that joint training on large image-text datasets can produce representations that support this type of retrieval across different domains.

Artwork retrieval presents a more challenging case for cross-modal alignment. Artworks often express abstract, symbolic, or stylistic meanings rather than concrete visual objects. At the same time, the associated texts tend to focus on historical context or interpretation instead of visible attributes. This mismatch makes it unclear whether models trained on general image-text data can align artworks and text effectively.

Because of this difficulty, it is important to examine semantic alignment in the art domain more carefully. Model architecture may influence how visual details

are represented. Textual input may also play a key role, since different forms of text provide different types of semantic information. Natural language descriptions often capture narrative meaning, while structured metadata provides categorical or temporal context.

This report studies cross-modal semantic alignment for artwork retrieval using the SemArt dataset. Rather than treating model choice and query design as fixed decisions, the analysis examines how these factors affect retrieval performance. The study first compares different representation models to identify a suitable embedding space. It then analyzes how different textual information sources influence retrieval under varying levels of difficulty.

Through these experiments, the report aims to clarify when semantic alignment works well for artworks and when it becomes more difficult. The analysis also aims to identify which combinations of visual representations and textual inputs support more reliable retrieval results.

2 Research Questions

In this report, I examine how effectively artworks and their associated texts can be aligned within a shared semantic embedding space, with a particular focus on retrieval-based evaluation. Rather than assuming a fixed representation model or query formulation, I explicitly examine how both model choice and textual information influence alignment quality. The study is guided by the following research questions.

Main Research Question. How effectively can visual and textual representations be aligned in a shared semantic space for artwork retrieval, and which factors influence the quality of this alignment?

This question captures the overall goal of the study. Alignment quality is examined through text-to-image retrieval performance using ranking-based evaluation metrics.

RQ1: Model Selection for Semantic Alignment. How does the choice of representation model affect the alignment between images and text in artwork retrieval?

This question motivates the comparison of different embedding models under the same retrieval setting. The goal is to assess whether certain cross-modal models provide stronger semantic alignment than others.

RQ2: Textual Information and Retrieval Difficulty. How do different forms of textual information, including natural language descriptions and structured metadata, influence retrieval performance under varying levels of difficulty?

This question focuses on the role of textual input in semantic alignment. It examines whether combining different text sources leads to more robust retrieval results, especially in more difficult cases.

3 Data

This study uses the SemArt dataset [1], which is designed for semantic understanding and cross-modal artwork retrieval. Each data instance includes an image of an artwork together with a natural language description.



Fig. 1: Example data instance from the SemArt dataset. Each artwork image is paired with a free-form textual description and structured metadata fields, including author, artistic type, school, and timeframe.

SemArt validation set. For model comparison, I use the SemArt validation split, which contains **1,069** image–text pairs. Each instance includes an artwork image, a free-form textual description, and associated metadata such as author, artistic type, school, and timeframe. The textual description typically provides a narrative interpretation of the artwork, while metadata fields encode categorical or temporal information that is not always visually explicit. Due to computational limitations, experiments are conducted on the validation split. The size of this split allows repeated evaluation across models and query construction strategies while remaining computationally tractable.

Human-defined easy and difficult subsets. To analyze retrieval behavior under different levels of semantic ambiguity, I further evaluate on two curated subsets labeled as *easy* and *difficult*. Each subset contains **100** queries, and for each query a fixed set of ten candidate images is provided, among which exactly one image is labeled as the ground truth. The easy subset consists of visually and semantically distinctive cases, whereas the difficult subset includes artworks with higher stylistic similarity or more abstract descriptions.

10-way retrieval setting. All retrieval experiments are conducted in a controlled 10-way setting. Given a textual query, models rank ten candidate images according to cosine similarity in the shared embedding space. Performance is evaluated based on the rank position of the ground-truth image. This setting enables a

focused analysis of semantic alignment and ranking behavior without the confounding effects of large-scale retrieval.

4 Methodology

This section describes the retrieval framework used to evaluate cross-modal semantic alignment. It introduces the embedding models, the query construction strategies, the retrieval procedure, and the evaluation metrics. All components are defined in a consistent way so that differences in performance can be attributed to representation quality and textual information, rather than to experimental design choices.

4.1 Cross-Modal Retrieval Framework

Given a textual query and a set of candidate images, both modalities are embedded into a shared semantic space. Text and image embeddings are ℓ_2 -normalized, and cosine similarity is used to measure semantic relatedness. For each query, candidate images are ranked in descending order of similarity, producing a ranked list in which higher positions indicate stronger semantic alignment.

This retrieval-based formulation allows semantic alignment to be evaluated indirectly through ranking quality: if visual and textual representations are well aligned in the embedding space, the correct image should appear at or near the top of the ranked list.

4.2 Representation Models

To study semantic alignment, I consider different representation models that map images and texts into embedding spaces.

CLIP. CLIP is a vision–language model trained on large-scale image–text pairs using contrastive learning. It jointly learns text and image encoders such that semantically corresponding inputs are close in the embedding space. Due to its cross-modal training objective, CLIP is a natural candidate for semantic alignment evaluation. In this study, both ViT-B/32 and RN50 variants are considered for model comparison.

ResNet50 baseline. As a vision-only baseline, ResNet50 features are used for image-to-image retrieval. This baseline does not provide cross-modal alignment, but serves as a sanity check to verify the integrity of the image processing pipeline and the discriminative power of visual features in isolation.

4.3 Query Construction

To analyze the contribution of different textual information sources, multiple query construction strategies are defined based on the available fields in the SemArt dataset:

- **Desc**: the natural language description of the artwork.
- **Meta**: structured metadata fields, including artistic type, school, timeframe, and technique.
- **Desc+Meta**: a concatenation of the description and metadata fields.
- **Structured**: a structured prompt that explicitly labels different fields (e.g., *School*:, *Timeframe*:, *Description*:).

These strategies allow a controlled comparison between free-form text, structured contextual information, and their combinations.

4.4 Retrieval Setting

All experiments are conducted in a controlled 10-way retrieval setting. For each query, exactly ten candidate images are provided, among which one image is labeled as the ground truth. Models rank the ten candidates based on cosine similarity to the query embedding. Retrieval-based evaluation is commonly used to assess the quality of cross-modal semantic embeddings [3].

The 10-way setting is chosen to isolate semantic alignment behavior under controlled conditions, enabling detailed analysis of ranking positions and distributional effects without the confounding factors present in large-scale retrieval.

4.5 Evaluation Metrics

Retrieval performance is evaluated using standard ranking-based metrics from information retrieval. Recall@k (R@k) measures the proportion of queries for which the ground-truth image appears within the top k ranked results. Mean Reciprocal Rank (MRR) measures the average inverse rank of the ground-truth image, which places more weight on higher ranking positions.

In addition to aggregate metrics, I analyze the distribution of ground-truth ranks to gain insight into how different query strategies affect early versus late ranking behavior, particularly under difficult retrieval conditions.

5 Experiment 1: Model Selection for Semantic Alignment

This experiment aims to compare different representation models in terms of their ability to align artworks and textual descriptions in a shared semantic space. Since subsequent analyses rely on the quality of this alignment, an explicit model selection step is required.

5.1 Motivation

Cross-modal retrieval performance is strongly influenced by the structure of the embedding space produced by the representation model. Although recent vision-language models have demonstrated strong performance on general benchmarks, their effectiveness in the art domain cannot be assumed. Therefore, I compare alternative representation models under identical retrieval conditions to justify the choice of a model for subsequent experiments.

5.2 Models Compared

I consider two cross-modal models and one vision-only baseline:

- **CLIP ViT-B/32**, a vision–language model using a Vision Transformer as the image encoder.
- **CLIP RN50**, a CLIP variant whose image encoder is based on a ResNet-50 backbone trained with a contrastive image–text objective.
- **ResNet50 (vision-only)**, a standard ResNet-50 model used for image-to-image retrieval as a sanity check.

The ResNet50 baseline is not expected to provide cross-modal alignment and is therefore included only to verify the integrity of the visual feature pipeline.

5.3 Experimental Setup

The experiment is conducted on the SemArt validation set, which contains 1,069 image–text pairs. For both CLIP variants, text-to-image retrieval is performed using the natural language description as the query. Image and text embeddings are ℓ_2 -normalized, and cosine similarity is used for ranking.

For the ResNet50 baseline, an image-to-image self-retrieval task is performed, where each image is used to retrieve itself from the candidate set. This setup serves as a sanity check rather than a direct comparison with cross-modal retrieval results.

Retrieval performance is evaluated using Recall@ k ($R@k$) and Mean Reciprocal Rank (MRR).

5.4 Results

As shown in Table 1, CLIP ViT-B/32 consistently outperforms CLIP RN50 across all retrieval metrics. The ResNet50 baseline achieves perfect self-retrieval accuracy in the image-to-image setting, confirming that the visual features are discriminative but not directly comparable to cross-modal retrieval performance.

Table 1: Text-to-image retrieval performance on the SemArt validation set (N=1,069).

Model	R@1	R@5	R@10	MRR
CLIP ViT-B/32	0.171	0.384	0.494	0.279
CLIP RN50	0.159	0.368	0.471	0.257

5.5 Model Selection

Based on the results of Experiment 1, CLIP ViT-B/32 is selected as the main representation model for the following analyses. This model shows stronger retrieval performance than CLIP RN50 across all evaluation metrics. The results

suggest that CLIP ViT-B/32 provides a more effective alignment between visual and textual representations in the shared semantic space. For this reason, it is used in all subsequent experiments that examine the role of textual information and retrieval difficulty.

6 Textual Information and Retrieval Difficulty

This section examines how different sources of textual information affect cross-modal semantic alignment under varying levels of retrieval difficulty. Using the representation model selected in Experiment 1, the analysis focuses on how query construction influences retrieval performance on both easy and difficult subsets.

6.1 Motivation

Even when the representation model is fixed, retrieval performance can vary depending on the content and structure of the textual query. In the art domain, different types of text provide different forms of semantic information. Natural language descriptions often express narrative or interpretative meaning. Structured metadata provides categorical, stylistic, or temporal context.

Because these information sources serve different roles, their effects on semantic alignment may also differ. This subsection examines how textual information influences retrieval performance, and whether its impact becomes more important as retrieval difficulty increases.

6.2 Experimental Setup

All experiments in this section use CLIP ViT-B/32, which is selected based on the results of Experiment 1. Evaluation is conducted on two curated subsets labeled as easy and difficult. Each subset contains 100 queries in a controlled 10-way retrieval setting.

Four query construction strategies are compared:

- **Desc**: natural language description only.
- **Meta**: structured metadata fields only.
- **Desc+Meta**: concatenation of description and metadata.
- **Structured**: a structured prompt explicitly labeling different fields.

For each query, candidate images are ranked by cosine similarity in the shared embedding space. Performance is evaluated using Recall@ k and Mean Reciprocal Rank (MRR).

6.3 Quantitative Results

Table 2 reports retrieval performance across query strategies and difficulty levels. On the easy subset, all strategies achieve relatively high performance, with

Table 2: Text-to-image retrieval performance under different query strategies and difficulty levels (10-way retrieval).

Split	Query	R@1	R@5	R@10	MRR
Easy	Desc	0.74	0.98	1.00	0.840
	Meta	0.45	0.94	1.00	0.648
	Desc+Meta	0.76	1.00	1.00	0.857
	Structured	0.80	0.99	1.00	0.886
Difficult	Desc	0.54	0.92	1.00	0.707
	Meta	0.22	0.70	1.00	0.436
	Desc+Meta	0.59	0.96	1.00	0.742
	Structured	0.56	0.95	1.00	0.713

the structured query yielding the highest Recall@1. On the difficult subset, performance drops across all strategies, reflecting increased semantic ambiguity. Notably, metadata alone performs poorly, while combining descriptions with metadata consistently improves Recall@1 and MRR compared to using descriptions alone.

6.4 Rank Distribution Analysis

To further analyze how metadata affects retrieval beyond aggregate metrics, I examine the distribution of ground-truth ranks on the difficult subset. Figure 2 shows the histogram of ground-truth ranks for the Desc and Desc+Meta strategies, while Figure 3 presents the corresponding cumulative distribution functions (CDFs).

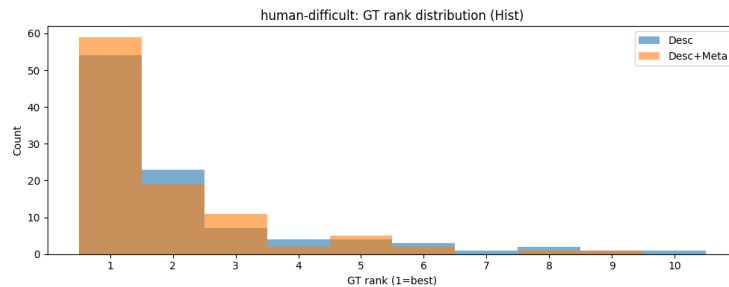


Fig. 2: Ground-truth rank histogram on the difficult subset for Desc and Desc+Meta queries.

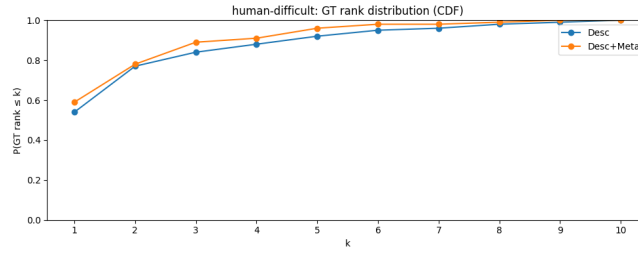


Fig. 3: Ground-truth rank CDF on the difficult subset for Desc and Desc+Meta queries.

The rank distribution analysis shows that augmenting descriptions with meta-data primarily increases the mass at top ranks, leading to higher probabilities of retrieving the correct image within the first few positions. However, the tail of the distribution remains largely unchanged, indicating that metadata refines early ranking rather than correcting severely misaligned cases.

6.5 Qualitative Analysis

To complement the quantitative findings, I analyze representative qualitative examples from the difficult subset. All qualitative examples are selected from the human-defined difficult subset. I select two cases where Desc+Meta improves the rank of the ground-truth image and two cases where performance degrades relative to Desc-only queries. These examples illustrate how metadata can help disambiguate stylistically similar artworks, but may also introduce noise when metadata fields are overly generic or weakly correlated with visual content.

Representative examples are shown in Figure 4 and Figure 5, with the ground-truth image highlighted.



Fig. 4: Qualitative difficult cases where metadata improves early retrieval ranks. The ground-truth image is highlighted.



Fig. 5: Qualitative examples from the difficult subset where adding metadata does not improve the rank of the ground-truth image compared to description-only queries. The ground-truth image is highlighted.

7 Discussion

This section examines the results from the two experiments and explains what they indicate about cross-modal semantic alignment in artwork retrieval. The discussion considers the effects of representation models and textual information, and it also uses retrieval difficulty to support the analysis.

7.1 Semantic Alignment and Model Choice

The results of Experiment 1 indicate that the choice of representation model affects retrieval performance. Both CLIP variants are trained using a cross-modal objective, yet CLIP ViT-B/32 shows consistently better retrieval performance than CLIP RN50. This result indicates that differences in visual encoder architecture affect how artistic features are represented and aligned with text.

7.2 Role of Textual Information

The results from Experiment 2 show that textual information strongly affects retrieval quality, especially under more difficult conditions. Natural language descriptions provide detailed semantic signals that are closely related to visual content. These descriptions alone already support relatively strong retrieval performance.

Structured metadata contributes a different type of information. Metadata often encodes stylistic, categorical, or temporal context that is not directly visible in the image. When metadata is combined with descriptions, retrieval performance improves, particularly at early ranking positions in the difficult subset.

This pattern suggests that metadata helps refine alignment when visual ambiguity is high.

At the same time, metadata alone performs poorly in fine-grained retrieval. This result indicates that structured fields lack the descriptive detail needed to distinguish between visually similar artworks. Metadata appears to be most useful when it complements, rather than replaces, natural language descriptions.

7.3 Retrieval Difficulty as an Analytical Tool

The comparison between the easy and difficult subsets highlights the value of retrieval difficulty for analysis. On the easy subset, most query strategies achieve high performance, which limits meaningful differences between approaches. In contrast, the difficult subset reveals clearer differences in ranking behavior across query constructions.

This contrast shows that difficult retrieval cases provide more information about semantic alignment quality. They make it easier to observe how additional textual context affects early ranking positions. For this reason, evaluating performance under challenging conditions offers more insight than relying on aggregate results alone.

7.4 Limitations

Several limitations should be noted. First, the experiments are conducted on the validation split only, which restricts the scale of evaluation. Second, the analysis uses a controlled 10-way retrieval setting, which does not fully reflect large-scale retrieval scenarios. Third, the study focuses on a single dataset and domain, which may limit generalization to other types of visual data.

Despite these limitations, the controlled design allows a clear examination of how representation models and textual information influence semantic alignment.

8 Conclusion

This report examined cross-modal semantic alignment in artwork retrieval using a controlled retrieval setting on the SemArt dataset. The study focused on how representation models and different forms of textual input influence retrieval behavior within a shared semantic space.

The analysis showed that the choice of representation model affects alignment quality, and that textual input plays an important role in retrieval performance. Natural language descriptions provided strong semantic information, while structured metadata contributed additional context when combined with descriptions. The effect of textual information became more visible under more difficult retrieval conditions.

The experiments were conducted under specific constraints, including a limited dataset split and a fixed retrieval setting. Within these limits, the results

help clarify how model design and textual input shape semantic alignment in the art domain.

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